

MOORHEAD, MN, USA

WMO: 722129

Lat: 46.839N

Lon: 96.663W

Elev: 280

StdP: 98.01

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-25.2	-31.1	0.2	-27.0	-28.0	0.3	-24.3	14.9	-13.6	13.3	-11.1	2.4	170	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.1	23.1	30.0	21.9	28.0	20.5	25.1	29.7	23.7	27.8	22.5	26.6	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.2	27.9	22.4	17.7	26.5	21.3	16.5	25.4	77.8	29.3	72.2	28.1	67.7	26.7	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	11.1	9.7	DB	-31.0	34.8	2.4	1.3	-32.7	35.7	-34.1	36.5	-35.4	37.2	-37.1	38.2	(4)
			WB	-31.2	27.2	2.2	1.6	-32.8	28.3	-34.1	29.3	-35.4	30.2	-37.0	31.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-11.8	-11.1	-2.1	6.4	13.9	19.5	21.7	20.0	16.2	8.3	-0.1	-9.0	(6)
(7)		DBStd	13.33	7.92	7.16	7.35	5.86	5.27	3.44	3.16	3.21	4.55	5.08	6.28	7.08	(7)
(8)		HDD10.0	2797	677	592	378	133	22	0	0	0	5	94	305	591	(8)
(9)		HDD18.3	4769	936	825	633	358	154	26	8	19	93	316	553	849	(9)
(10)		CDD10.0	1368	0	0	4	26	144	286	364	310	192	40	2	0	(10)
(11)		CDD18.3	298	0	0	0	1	18	62	113	71	30	3	0	0	(11)
(12)		CDH23.3	2721	0	0	0	8	222	531	1052	603	272	32	0	0	(12)
(13)		CDH26.7	786	0	0	0	1	74	136	332	161	75	7	0	0	(13)
(14)	Wind	WSAvg	4.2	4.4	4.4	4.6	4.9	4.1	3.2	3.2	3.8	4.6	4.5	4.3	(14)	
(15)	Precipitation	PrecAvg	612	16	16	29	44	82	111	82	66	56	22	22	(15)	
(16)		PrecMax	833	38	38	92	93	192	243	206	184	161	134	98	49	(16)
(17)		PrecMin	412	0	3	1	8	34	28	24	23	8	1	0	5	(17)
(18)		PrecStd	112	11	9	18	23	42	54	45	32	38	35	22	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	7.6	19.1	23.9	32.4	32.6	33.8	32.6	32.2	27.4	17.3	7.6	(19)
(20)		MCWB	2.8	5.3	14.5	13.0	17.9	22.2	25.4	23.9	21.8	17.3	11.5	5.9	(20)	
(21)		2%	DB	2.4	2.9	15.0	21.2	28.6	29.9	32.1	30.2	28.8	22.6	13.0	3.7	(21)
(22)		MCWB	1.3	1.7	11.0	12.1	17.0	20.0	24.6	23.0	20.2	14.6	8.5	1.8	(22)	
(23)		5%	DB	1.2	2.0	11.2	17.9	26.0	27.8	30.0	27.9	27.0	18.8	11.1	2.1	(23)
(24)		MCWB	0.5	1.1	8.2	10.6	16.0	19.1	23.3	21.3	19.1	13.2	7.9	1.0	(24)	
(25)		10%	DB	-1.0	-0.9	7.4	16.2	22.7	27.0	28.0	27.1	23.9	16.2	7.8	0.2	(25)
(26)		MCWB	-1.2	-1.3	5.6	9.5	14.6	18.5	21.6	20.5	18.0	11.0	5.1	-0.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	5.5	16.1	15.6	21.7	24.9	27.6	25.6	23.7	18.2	13.2	6.0	(27)
(28)		MCDWB	4.3	7.7	18.2	21.2	26.1	29.8	30.7	29.9	28.1	24.0	17.0	7.7	(28)	
(29)		2%	WB	1.4	2.2	11.2	13.2	19.5	22.9	25.7	24.3	22.1	15.8	9.8	2.3	(29)
(30)		MCDWB	2.2	3.0	14.3	18.6	24.9	27.2	30.4	28.3	27.3	21.5	12.6	3.3	(30)	
(31)		5%	WB	0.4	0.6	7.9	11.7	17.8	21.4	24.3	23.1	20.4	13.8	7.6	0.9	(31)
(32)		MCDWB	1.1	1.5	10.4	16.7	22.4	25.5	28.5	26.8	24.4	17.8	10.3	1.8	(32)	
(33)		10%	WB	-1.1	-1.2	5.4	10.2	16.3	20.1	23.0	21.8	19.1	11.8	5.4	-0.5	(33)
(34)		MCDWB	-0.8	-0.7	8.0	15.0	20.6	24.2	27.0	25.4	23.3	15.1	7.9	0.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.9	9.1	8.9	10.4	11.5	10.5	11.1	11.3	11.4	9.5	8.4	8.1	(35)	
(36)		5% DB	MCDBR	9.7	9.5	13.3	15.0	15.3	12.7	12.6	12.8	14.1	14.2	11.9	10.3	(36)
(37)		MCWBR	8.8	8.2	9.1	8.1	7.3	6.1	6.5	6.5	6.8	7.8	7.9	8.7	(37)	
(38)		5% WB	MCDBR	9.4	9.5	12.2	12.8	11.8	10.2	11.1	11.1	11.7	12.4	11.3	9.2	(38)
(39)		MCWBR	8.6	8.4	8.8	7.8	6.9	6.4	6.8	6.6	6.9	7.4	8.0	8.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.303	0.306	0.360	0.378	0.391	0.392	0.395	0.359	0.317	0.287	0.276	(40)	
(41)		taud	2.227	2.276	2.426	2.333	2.347	2.352	2.345	2.335	2.421	2.516	2.503	2.379	(41)	
(42)		Ebn at Noon	769	861	924	894	887	874	868	849	852	846	811	775	(42)	
(43)		Edh at Noon	82	96	97	117	120	120	120	116	97	76	63	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.66	2.81	3.87	4.91	5.50	6.12	6.42	5.39	4.03	2.52	1.56	1.18	(44)	
(45)		RadStd	0.17	0.30	0.39	0.40	0.37	0.41	0.26	0.45	0.26	0.36	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page